ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Technology (CSE)

QUIZ #1 (SET-A) **SUMMER SEMESTER, 2023-2024**

DURATION: 30 MINUTES **FULL MARKS: 15**

CSE 4711: Artificial Intelligence

Programmable calculators are not allowed.

Answer **all 2 (two)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

Student ID:

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1. The Tower of Hanoi is a classic puzzle which consists of three rods (A, B, C) and n disks of different sizes, which can slide onto any of the rods. The puzzle starts with the all of the disks neatly stacked in decreasing order of size (largest at the bottom, smallest at the top) on rod A. The objective is to move the entire stack to rod C, obeying the following rules: 6 (CO1) (PO1)

   - Only one disk can be moved at a time.
   - Only the topmost disk from any rod can be moved.
   - No disk can be placed on top of a smaller disk.

   Formulate the problem as a single-agent search by identifying the state space, start state, actions, successor function, goal test, and cost for $n = 3$.

Solution:

- **State space:** Each state consists of a collection of three sets (one for each rod) and each set contains the disks in order from top to bottom. The disks are represented by a number from 1 to 3, where the value determines the size of the disk. One possible state could be: ({2, 3}, {1}, {}), where the A has two disks (2 and 3), B has only one (1), and C has none.
- **Start state:** ({1, 2, 3}, {}, {}).
- **Actions:** An action is defined as moving the top disk from one rod to another only if the source rod is not empty and the destination rod is empty or its top disk is larger than the disk being moved.
- **Successor function:** Takes the current state and a valid action to move one disk from a rod to another.
- **Goal test:** Check if current state is ({}, {}, {1, 2, 3}).
- **Cost:** Number of moves made.

Rubric:

- 1 point for each correctly answered segment.

2. Consider a state space where the states are represented by integers, the start state is the number 1 and each state k has two successors: $2k$ and $2k+1$. If the goal state is 11, list the order in which the nodes will be visited for: 9 (CO1) (PO1)

1. Breadth-first search
2. Depth-limited search with limit = 3
3. Iterative deepening search with limit = 3

Solution:

- **Breadth-first search:** 1 2 3 4 5 6 7 8 9 10 11
- **Depth-limited search:** 1 2 4 8 9 5 10 11
- **Iterative deepening search:** 1; 1 2 3; 1 2 3 4 5 6 7; 1 2 4 8 9 5 10 11

Rubric:

- 3 points for each correct answer.